

Cost Benefit Analysis

Cost-benefit analysis (CBA) is a systematic approach used to evaluate the financial implications of a decision or project by comparing its projected costs and benefits. The goal is to determine if the potential benefits outweigh the costs, helping decision-makers make informed choices.

Cost-benefit analysis formulas

There are three formulas to help perform a basic cost-benefit analysis: The present value (PV) formula, the net present value (NPV) formula, and the cost-benefit analysis (CBA) ratio.

To calculate the CBA ratio, you must first use the PV formula to find the present value of the costs and the benefits of the project that's being analyzed.

$$\text{Present Value (PV)} = \text{FV} / (1+r)^n$$

Where:

- PV = Present Value
- FV = Future Value
- r = Discount Rate
- n = Number of periods

Apply this formula to get the PV of future costs and PV of future benefits.

Then, insert those values into the NPV formula and CBA ratio, as seen below:

$$\text{NPV} = \text{PV of Benefits} - \text{PV of Costs}$$

$$\text{CBA ratio} = \text{PV of Benefits} / \text{PV of Costs}$$

If the NPV is a positive number and the CBA ratio is greater than 1, that means the present value of the benefits outweighs the present value of the costs. Therefore, the project being analyzed is financially feasible.

Cost-benefit analysis (CBA) example with formulas

Example 1: Let's consider a hypothetical example of a CBA for a company deciding whether to invest in new manufacturing equipment:

Project: Purchase of new manufacturing equipment

- Cost of equipment: \$500,000
- Expected useful life: 5 years
- Annual maintenance cost: \$20,000
- Expected annual increase in production: 10,000 units
- Selling price per unit: \$50
- Variable cost per unit: \$30
- Discount rate: 10%

Step 1: Calculate potential annual financial benefits

Annual gross profit = 10,000 units x (\$50 selling price - \$30 cost) = \$200,000

Annual net profit = \$200,000 gross profit - \$20,000 maintenance = \$180,000

The potential annual financial benefit of new manufacturing equipment is \$180,000 per year.

Step 2: Calculate present value of financial benefits

To calculate the total present value (PV) for the potential financial benefits, we use the expected useful life of the new equipment (five years) as our 'r' value in the PV formula and calculate PV for each year.

Year 1 PV of benefits = \$180k / (1+10%)¹ = 163,636.36

Year 2 PV of benefits = \$180k / (1+10%)² = 148,760.24

Year 3 PV of benefits = \$180k / (1+10%)³ = 135,164.73

Year 4 PV of benefits = \$180k / (1+10%)⁴ = 122,084.24

Year 5 PV of benefits = \$180k / (1+10%)⁵ = 111,024.36

Finally, we sum the PV of each of the next five years to get the total PV of the potential financial benefits of this project.

Total PV of Benefits = \$680,669.93

Step 3: Calculate present value of costs

Next, we need to calculate the total present value of costs by adding the upfront cost (\$500,000) to the present value of annual maintenance costs. To calculate the present value of annual maintenance costs, we'll use the PV formula again.

Year 1 PV of costs = \$20k / (1+10%)¹ = 18,181.82

Year 2 PV of costs = \$20k / (1+10%)² = 16,528.93

Year 3 PV of costs = \$20k / (1+10%)³ = 15,029.77

Year 4 PV of costs = \$20k / (1+10%)⁴ = 13,658.69

Year 5 PV of costs = \$20k / (1+10%)⁵ = 12,345.68

Now we sum up these present values for the PV of maintenance costs.

PV of Annual Maintenance Costs = \$75,744.89

Finally, we can calculate the total present value of costs, which includes both the initial cost of equipment (\$500,000) and the present value of maintenance costs (\$75,744.89).

$$\text{Total PV of Costs} = 500,000 + 75,744.89 = \$575,744.89$$

Step 4: Calculate NPV and CBA ratio

Finally, we'll insert our calculated values into the NPV formula and CBA ratio.

$$\text{NPV} = \text{PV of Benefits} - \text{PV of Costs}$$

$$\text{Net Present Value (NPV)} = \$680,669.93 - \$575,744.89 = \$104,925.04$$

$$\text{Since, CBA ratio} = \text{PV of Benefits} / \text{PV of Costs}$$

$$\text{CBA Ratio} = \$680,669.93 / \$575,744.89 = 1.18$$

In this example, the positive NPV and benefit-cost ratio greater than 1 suggest that the investment in new equipment is financially viable.

Example 2:

Imagine a company, "TechSolutions," is considering launching a new mobile app.

- **Costs:**

- Development: \$50,000
- Marketing: \$20,000
- Software licenses: \$10,000
- Total costs: \$80,000

- **Benefits:**

- Increased revenue from app sales and subscriptions: \$150,000
- Reduced customer support costs: \$10,000
- Enhanced brand reputation: (estimated at \$20,000, a subjective value)
- Total benefits: \$180,000

Net Present Value (NPV):

Assuming a 10% discount rate and a 3-year project life, the NPV calculation would show that the total present value of benefits exceeds the total present value of costs.

Example 3

Here is an example of a company exploring a new equipment purchase: Speed Inc, a regional delivery company, is considering purchasing a new fleet of vehicles for its staff to improve fuel efficiency and storage space, which it hopes can allow drivers to deliver more products at a lower cost. The company wants to know whether this new equipment might save it money in the long run, or whether it might increase its costs, so it decides to perform a cost-benefit analysis. To do this, management determines the new vehicles would cost the company \$100,000. After factoring in gas mileage, maintenance and storage, the company estimates the new vehicles can save it \$138,000 per year. Using these figures, the company performs a CBA and determines its savings from purchasing the new vehicles outweigh the cost of buying them, and management decides to finalize the purchase.

Here's the formula it used to make this decision:

Value = benefits – costs

Value = 138,000 - 100,000

Value = 38,000

Example 4

Here's an example of a company deciding whether to update its website to increase sales: Threads and Kicks, an apparel company, has noticed its sales have steadily decreased over the past few months. After surveying customers, management staff learns some people think the company's website is slow and outdated, which makes it challenging to use. To counter this, management uses a CBA to determine whether the costs of updating the site will increase sales enough to make it a worthwhile project. Counting labor, time and upkeep, the company determines it may cost them \$68,000 per year to update and manage the new site. Its finance team, through its research and more customer feedback, estimates an improved website can help the company make \$5,000 per month or \$60,000 per year. With

these figures, management decides updating the website isn't beneficial now, and they decide to wait a few more months before considering it again.

Here's the formula they used for their CBA:

Value = benefits – costs

Value = 60,000 - 68,000

Value = -8,000